uOttawa

Project 1: Emotion Detection (27.5%)

Due: *April 12th at 11:59PM*

This is a group Project.

(Team leader should submit on behalf of the team on Brightspace)

Dataset

The SMILE_PLUS dataset contains facial images of subjects presenting a "happy" or "neutral" emotional expression. Moreover, the dataset package contains two annotation files: ***"annotations.arff"*** and ***"annotations.csv"***. The annotation files describe the emotional expression presented in each image. The file ***"annotations.arff"*** is compatible with Weka, a machine learning tool that you can use for this assignment. You can use the generic ***"annotations.csv"*** file if you decide to use any other machine learning tool or library.

Machine Learning Tool or Library

You can use any machine learning tool or library for this project. If you are new to machine learning, we recommend that you use Weka. Weka is an open-source tool that implements several machine learning algorithms. You can use the Weka Package Manager to download additional modules for feature extraction and classification. Please refer to the ***"Weka Installation Guide"*** for instructions regarding the setup of Weka on your computer.

Problem

Develop a machine learning solution to address the binary classification problem of distinguishing between the "neutral" and "happy" facial expressions presented in the SMILE dataset.

To test the performance of your model, perform a cross-validation with 10 folds. Report the following performance measures on the cross-validation:

- False Positive Rate
- Precision
- Recall
- F-Measure
- Confusion Matrix

Consider "happy" as the positive class in your performance measures' calculation.

The instructor will evaluate your model on an unpublished testing set. Your system will primarily be evaluated based on the F-Measure. Therefore, consider it your most important performance metric.

What to Submit

Submit a document that describes your work in sufficient detail so that your results can be reproduced by the reader. The document should have the following sections:

- **Tool(s)**: Provide a concise description of the machine learning tool(s) or library/libraries you used.
- **Features:** Describe the features you used. If you use a feature extraction technique not discussed in class, describe it in detail. Explain how you extracted your features (i.e. which tool(s), library/libraries, or algorithm(s) you used to obtain your features).
- **Classification:** Specify which classifier(s) you used and whether you applied an ensemble-learning scheme. Describe the process through which you arrived at your design. Describe any additional schemes you might have employed (e.g. dimensionality reduction). If you used a classifier not discussed in class, describe it in detail (i.e. how it works).
- **Hyperparameters**: Specify all the hyperparameters you tuned. Describe what these hyperparameters are and what is their effect is on the operation of the classifier. Describe your hyperparameter tuning process.

If you used a programming library (e.g., scikit-learn, tensor flow), you must submit your code along with instructions on how to execute it.

Other Considerations

Think Outside of the Classroom Box

You do not need to rely solely on the concepts we learned in class. You may try classifiers and image filters that we have not seen in class. You can try out creative ensemble configurations. Maybe you can learn about dimensionality reduction or hyperparameter optimization (e.g. grid search) to improve performance.

Deep Learning

Given the small size of the dataset, training a deep network from scratch might not render good performance. However, you can try to apply transfer learning to fine-tune a pre-trained network.